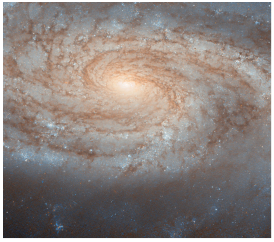
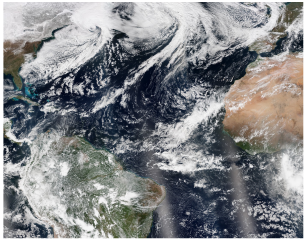
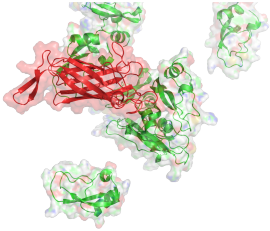
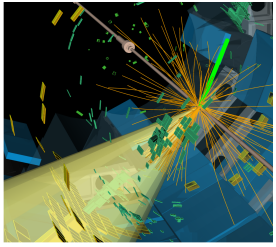

High performance scientific computing in C++

HPC C++ Course 2026

12 October – 15 October 2026 | Sandipan Mohanty | Forschungszentrum Jülich, Germany

Member of the Helmholtz Association

JÜLICH Forschungszentrum

Course material

- To download the material, use git, or the web browser. Here is the git repo: https://gitlab.jsc.fz-juelich.de/sdlbio-courses/hpcxx2026.git

```
cd your_working_directory
git clone https://gitlab.jsc.fz-juelich.de/sdlbio-courses/hpcxx2026.git
```

  The material will be updated every day during the course.

- Logging in to our system as described in the following slides and running the `setup.sh` script sets up your working directory for the course. Please read the information printed when you run the `setup.sh` script. It is important to work without interfering with others. Please download and work on the course material in **your own directory!**

High performance scientific computing in C++

Supercomputer access for the exercises

- Please login to the Jupyter-JSC system (https://jupyter-jsc.fz-juelich.de). If asked to choose whether to use JSC account or Helmholz account, choose JSC.
- After logging in, try to add a new jupyterlab. Choose JURECA as the system and training2652 as the project.
- For the partition choose **LoginNode**, and then start. Wait until the swirly things stop and you see the panel.
- What we will most need from there is the terminal, which should be at the bottom.
- In the terminal type this:

```
$ source $PROJECT/local/setup.sh
```

- After this, your paths should be set correctly. Test it using

```
$ g++ --version
$ clang++ --version
```

You should see GCC version 16.2 and Clang version 23.1.0, both compiled from source.

High performance scientific computing in C++

- **Necessary first step:** `source $PROJECT/local/setup.sh`, for every new terminal on JURECA for this course.
- It creates user specific working directories, downloads and updates course material and sets up the environment variables for compilers and libraries.
- After the script `setup.sh` is sourced, the following environment variables (EV) and additional shortcuts (SC) are available
  - `cxx2026`: (EV) Location of your private working area for the course (like `$HOME`)
  - `swhome`: (EV) Top level folder for software installations for compilers and libraries
  - `cdp`: (SC) Change directory to the top level of your private workspace (≡ `cd $cxx2026`)
  - `pathadd`: (SC) Prepend a new folder to PATH. E.g., `pathadd /x/y/z/bin`
  - `pathrm`: (SC) Remove a folder from PATH
  - `libpathadd`, `libpathrm`: (SC) Same as above, but for `LD_LIBRARY_PATH`, `LD_RUN_PATH`, `LIBRARY_PATH`
  - `incpathadd`, `incpathrm`: (SC) Same, but for `CPATH`, which is searched by the compilers for include files.
  - `cmpathadd`, `cmpathrm`: Same, but for `CMAKE_PREFIX_PATH`
  - `G`: (SC) Alias for `g++` using common options `-std=c++23 -pedantic -Wall -O3 -march=native`
  - `C`: (SC) Similar to `G`, but for Clang. It also uses Clang's own implementation of the standard library, `libc++`
  - `Gstdm` (SC). Generates pre-compiled module for the standard library, which can then be imported with a simple "import std;" line. For every new folder where you need to compile programs using "import std;" using the g++ compiler, you need to run this once. Alias for
    `g++ -std=c++23 -fmodules -fsearch-include-path -c bits/std.cc`

Some notes on the organisation of the course material

- When you run
  source $PROJECT/local/setup.sh
- A working directory with your user name will be created (if it was not already there) inside the project working area
- Type `cdp` to change directory to your private working directory.
  - Download course material in that directory as described earlier
  - Create a different folder, (e.g., `work`), where you do your modifications. Copy any examples you want to work on, to this directory, leaving the original archive unedited. How you manage things in `work` is up to you.
- The contents of these first few slides, since you may need to look them up later, are placed in the file `utilities.pdf`
- To test if things are working fine for you, go into the folder `test_your_setup` (or copy that into your working directory), and follow the instructions in the README files there. Finishing these steps ahead of the course will help save some time during the course!

Direct SSH connection to JURECA

- This is not strictly necessary for the course, but many people prefer to SSH connections outside the Jupyter session. This is also a useful backup option if Jupyter-JSC has an outage during the course!
- Follow the instructions in the page https://apps.fz-juelich.de/jsc/hps/jureca/clus ter/access.html#ssh-login to generate a suitable SSH key for JURECA. Certain key types are preferred, and the recommended key type is ed25519.

```
$ ssh-keygen -a 100 -t ed25519 -f ~/.ssh/id_ed25519_jsc
$
```

- Upload the generated **public key** to JuDoor
  - Login to JuDoor
  - Click on "Manage SSH-keys" next to JURECA
  - Select your ed25519 public key file using the "Browse" button. This file has a ".pub" extension.
  - Pay special attention to the "from clause". This goes in the box to the right of the public key filename box. If you are in a hurry, use the suggestion based on your current IP address, which is displayed in the from-clause box by pressing the downward arrow key to its right.
  - After filling in a suitable "from clause", click on the Start upload of SSH-keys button
- Wait

Direct SSH connection to JURECA

- Open SSH session on JURECA using your ed25519 key like this:

```
$  ssh -i ~/.ssh/id_ed25519_jsc <yourid>@jureca.fz-juelich.de
```

- ... or, make it a little easier for yourself:
  - Add a section for JURECA in your SSH config file `~/.ssh/config`

```
1  Host jureca??
2    Hostname %h.fz-juelich.de
3    Match Host jureca.fz-juelich.de,jureca??.fz-juelich.de
4    User yourid
5    IdentityFile ~/.ssh/id_ed25519_jsc
6    ServerAliveInterval 60
```

  - Open SSH session like this: `$ ssh jureca`
- After opening a new SSH session, set your project to training2652 and run the setup script:

```
$ jutil env activate -p training2652
$ source $PROJECT/local/setup.sh
$
```